IBM University Engagement Project Submission

“Exploring the Power of Machine Learning with IBM watsonx”

Project Tile – Power System Fault Detection and Classification

Domain of Project – Machine Learning

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Shivam Dhamankar		shivamdhamankar121@gmail.com	8788138398

Problem Statement -

Problem Statement : Power System Fault Detection and Classification

The Challenge

Design a machine learning model to detect and classify different types of faults in a power distribution system. Using electrical measurement data (e.g., voltage and current phasors), the model should be able to distinguish between normal operating conditions and various fault conditions (such as line-to-ground, line-to-line, or three-phase faults). The objective is to enable rapid and accurate fault identification, which is crucial for maintaining power grid stability and reliability.

The Objective

- Collect and process electrical system data from power distribution networks.
- Detect abnormal operating conditions and identify the occurrence of faults in real time.
- Classify different fault types, including:
  - Line-to-Ground (LG) Fault
  - Line-to-Line (LL) Fault
  - Double Line-to-Ground (LLG) Fault
  - Three-Phase (LLL) Fault
- Train and deploy machine learning models using IBM Watson Studio and IBM Cloud services for accurate fault prediction.
- Improve fault detection speed and classification accuracy compared to traditional methods.
- Reduce power interruptions, equipment damage, and maintenance costs through early fault identification.
- Enhance the reliability, stability, and operational efficiency of power distribution systems.
- Provide a scalable cloud-based solution that can support real-time monitoring and intelligent decision-making in smart grid environments.

Technology Used

Cloud Platform
IBM Cloud
IBM Watsonx Studio
IBM Watson Machine Learning
IBM Cloud Object Storage

Programming Language
Python

Machine Learning Libraries
Scikit-learn
Pandas
NumPy

Data Processing & Analysis
Data Preprocessing
Feature Engineering
Data Normalization

Dataset Format
CSV Files

Development Environment
Jupyter Notebook (IBM Watson Studio)

Proposed Solution -

Proposed Solution: Power System Fault Detection and Classification
The proposed system utilizes Machine Learning and IBM Cloud services to develop an intelligent Power System Fault Detection and Classification solution. The system collects electrical measurement data, including voltage and current phasors, from the power distribution network. This data is stored securely in IBM Cloud Object Storage for further processing and analysis.

The collected data is pre-processed to remove noise, handle missing values, and normalize measurements. Relevant features are extracted and used to train a machine learning model in IBM Watson Studio. The model is trained using historical power system data containing both normal operating conditions and various fault scenarios.

The system is designed to identify and classify different types of faults such as Line-to-Ground (LG), Line-to-Line (LL), Double Line-to-Ground (LLG), and Three-Phase (LLL) faults. Advanced machine learning algorithms are employed to achieve high prediction accuracy and reliable performance.

Once trained, the model is deployed using IBM Watson Machine Learning services. During real-time operation, incoming electrical measurements are analyzed by the deployed model to detect the occurrence of faults instantly. The system then classifies the fault type and generates alerts for operators.

A monitoring dashboard provides visualization of fault conditions and system status, enabling quick decision-making. The cloud-based architecture ensures scalability, accessibility, and efficient data management. This solution helps reduce fault detection time, minimize equipment damage, decrease power outages, improve maintenance efficiency, and enhance the overall reliability and stability of the power distribution system. It supports modern smart grid applications by providing automated, accurate, and real-time fault monitoring capabilities.

Component Used -

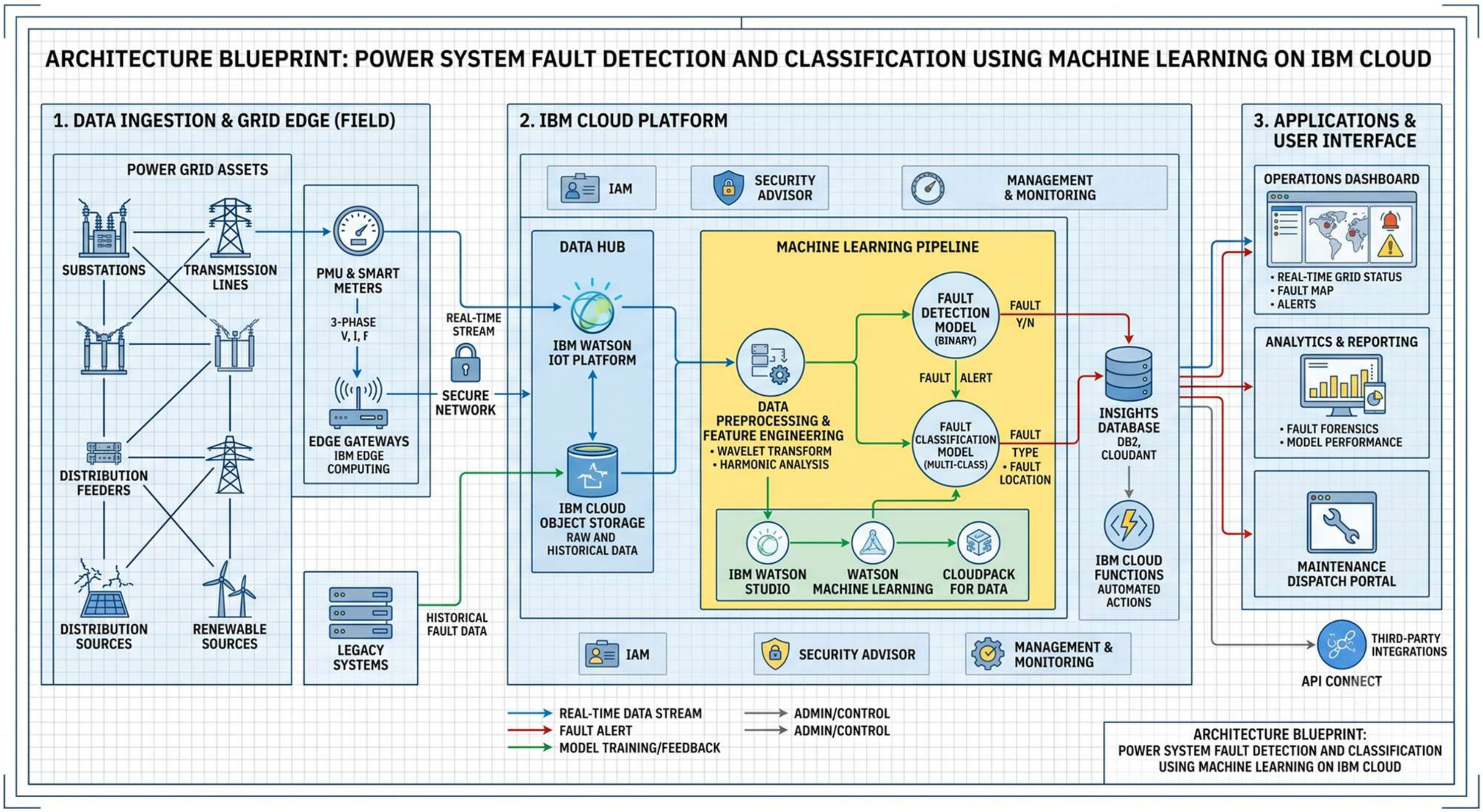
Hardware Components
Computer/Laptop
Power System Dataset (Voltage & Current Measurements)
Internet Connection

Software Components
IBM Cloud
IBM Watson Studio
IBM Watson Machine Learning
IBM Cloud Object Storage
Jupyter Notebook
Python

Machine Learning Components
Data Preprocessing Module
Feature Extraction Module
Model Training Module
Fault Detection Module
Fault Classification Module
Model Deployment Module

Output Components
Real-Time Fault Prediction
Monitoring Dashboard
Fault Alerts and Reports

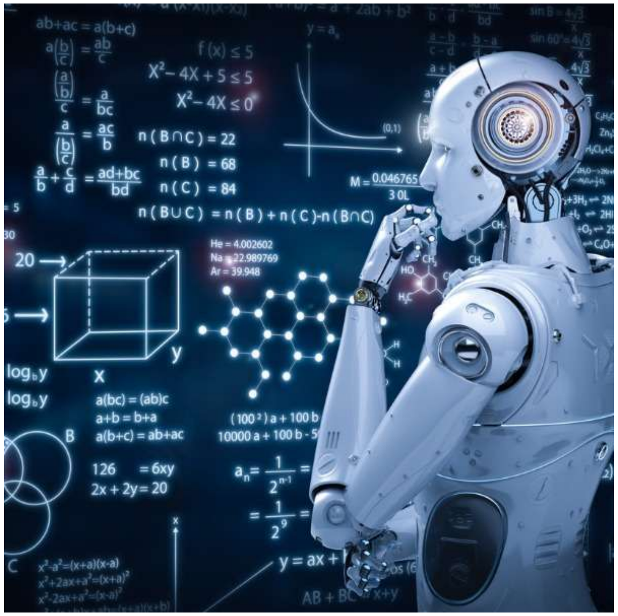
Architecture Blueprint



Role of Machine Learning in the solution

Machine Learning plays a key role in the proposed Power System Fault Detection and Classification system. It enables the automatic analysis of electrical measurement data, such as voltage and current phasors, to identify abnormal conditions in the power network. The ML model learns patterns from historical data and distinguishes between normal operating conditions and different fault types. Once trained, the model can accurately classify faults such as Line-to-Ground (LG), Line-to-Line (LL), Double Line-to-Ground (LLG), and Three-Phase (LLL) faults.

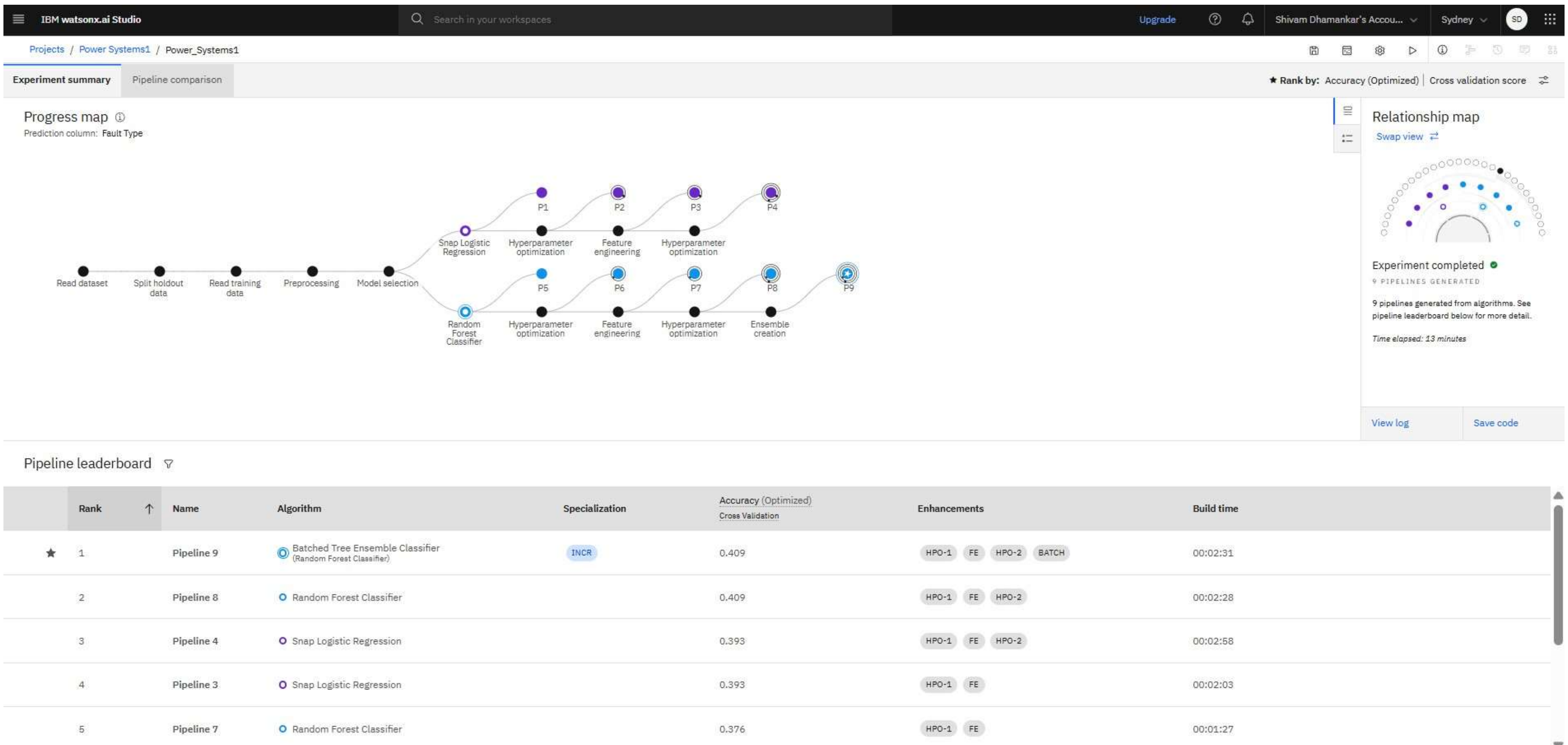
Machine Learning improves the speed and accuracy of fault detection compared to traditional rule-based methods. It reduces human intervention, supports real-time decision-making, and helps operators respond quickly to system failures. By providing rapid and reliable fault classification, ML enhances power grid stability, minimizes equipment damage, reduces downtime, and improves the overall reliability and efficiency of the power distribution system.



Technology Used

- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **IBM Watsonx.ai -** For model deployment, embeddings, and AI governance
- **Programming Language -** Python
- **Machine Learning Libraries**
- **Data Processing & Analysis**
- **Development Environment**
  Jupyter Notebook (IBM Watson Studio)

Results :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Sydney
SD
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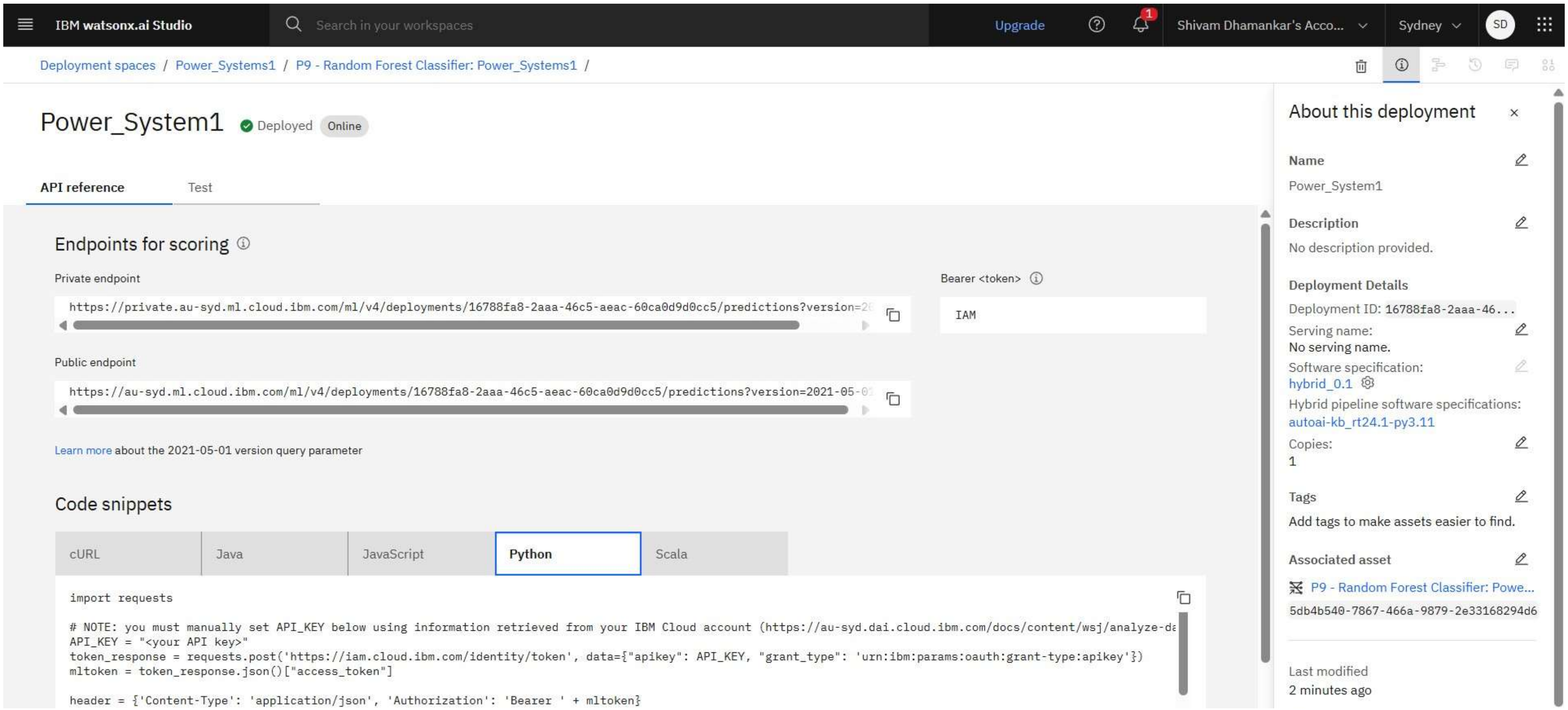
Experiment summary
Pipeline comparison
★ Rank by: Accuracy (Optimized) | Cross validation score
Progress map
Prediction column: Fault Type
Read dataset
Split holdout data
Read training data
Preprocessing
Model selection
Snap Logistic Regression
Random Forest Classifier
Hyperparameter optimization
Feature engineering
Hyperparameter optimization
Ensemble creation
P1
P2
P3
P4
P5
P6
P7
P8
P9
Relationship map
Swap view
Experiment completed
9 PIPELINES GENERATED
9 pipelines generated from algorithms. See pipeline leaderboard below for more detail.
Time elapsed: 13 minutes
View log
Save code
Pipeline leaderboard
Rank
Name
Algorithm
Specialization
Accuracy (Optimized)
Cross Validation
Enhancements
Build time
1 Pipeline 9 Batched Tree Ensemble Classifier (Random Forest Classifier) INCR 0.409 HPO-1 FE HPO-2 BATCH 00:02:31
2 Pipeline 8 Random Forest Classifier 0.409 HPO-1 FE HPO-2 00:02:28
3 Pipeline 4 Snap Logistic Regression 0.393 HPO-1 FE HPO-2 00:02:58
4 Pipeline 3 Snap Logistic Regression 0.393 HPO-1 FE 00:02:03
5 Pipeline 7 Random Forest Classifier 0.376 HPO-1 FE 00:01:27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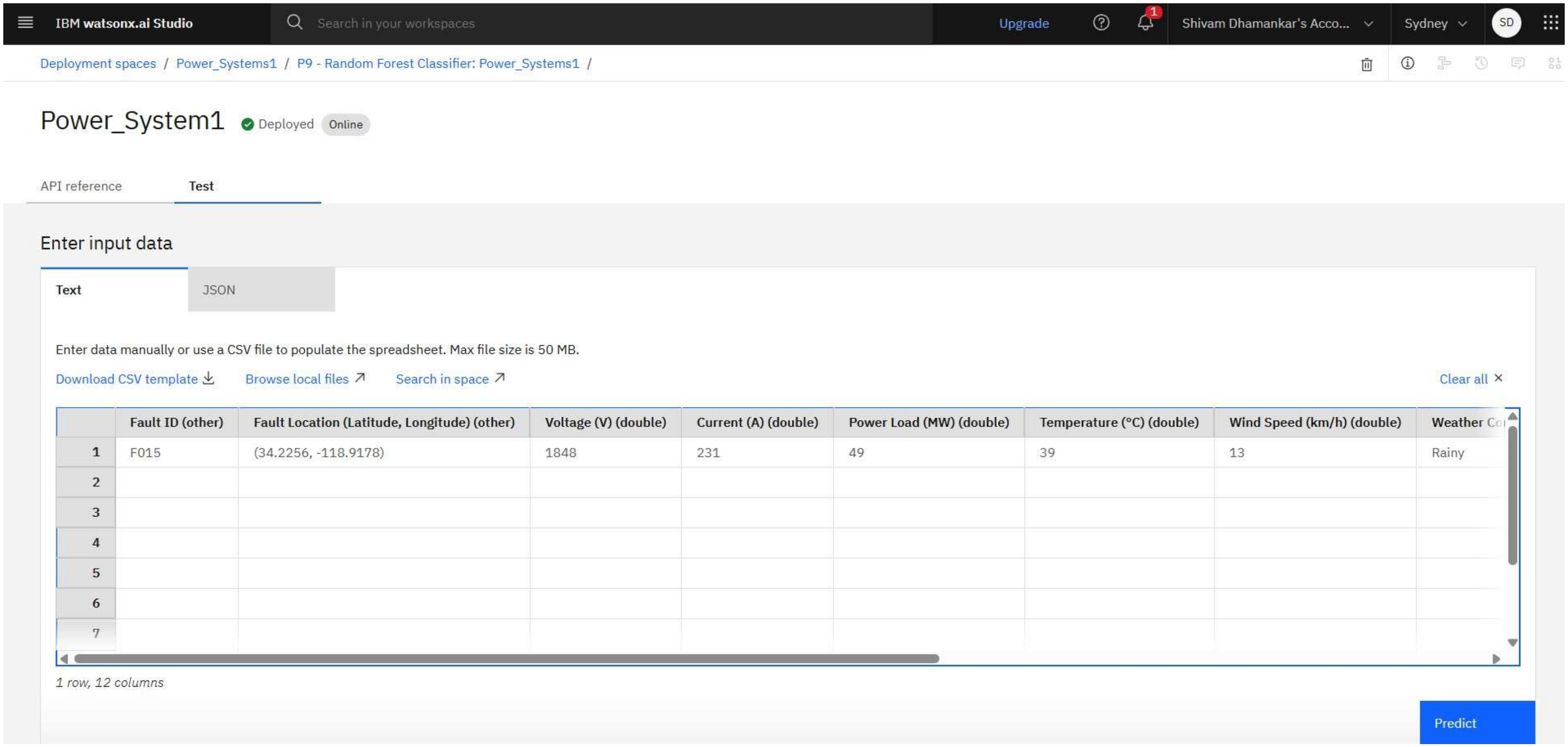
SD
Deployment spaces / Power_Systems1 / P9 - Random Forest Classifier: Power_Systems1 /
Power_System1
Deployed
Online
API reference
Test
Endpoints for scoring
Private endpoint
https://private.au-syd.ml.cloud.ibm.com/ml/v4/deployments/16788fa8-2aaa-46c5-aeac-60ca0d9d0cc5/predictions?version=
Bearer <token>
IAM
Public endpoint
https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/16788fa8-2aaa-46c5-aeac-60ca0d9d0cc5/predictions?version=2021-05-0
Learn more about the 2021-05-01 version query parameter
Code snippets
cURL
Java
JavaScript
Python
Scala
import requests
# NOTE: you must manually set API_KEY below using information retrieved from your IBM Cloud account (https://au-syd.dai.cloud.ibm.com/docs/content/wsj/analyze-da
API_KEY = "<your API key>"
token_response = requests.post('https://iam.cloud.ibm.com/identity/token', data={"apikey": API_KEY, "grant_type": 'urn:ibm:params:oauth:grant-type:apikey'})
mltoken = token_response.json()["access_token"]
header = {'Content-Type': 'application/json', 'Authorization': 'Bearer ' + mltoken}
About this deployment
Name
Power_System1
Description
No description provided.
Deployment Details
Deployment ID: 16788fa8-2aaa-46...
Serving name:
No serving name.
Software specification:
hybrid_0.1
Hybrid pipeline software specifications:
autoai-kb_rt24.1-py3.11
Copies:
1
Tags
Add tags to make assets easier to find.
Associated asset
P9 - Random Forest Classifier: Powe...
5db4b540-7867-466a-9879-2e33168294d6
Last modified
2 minutes ago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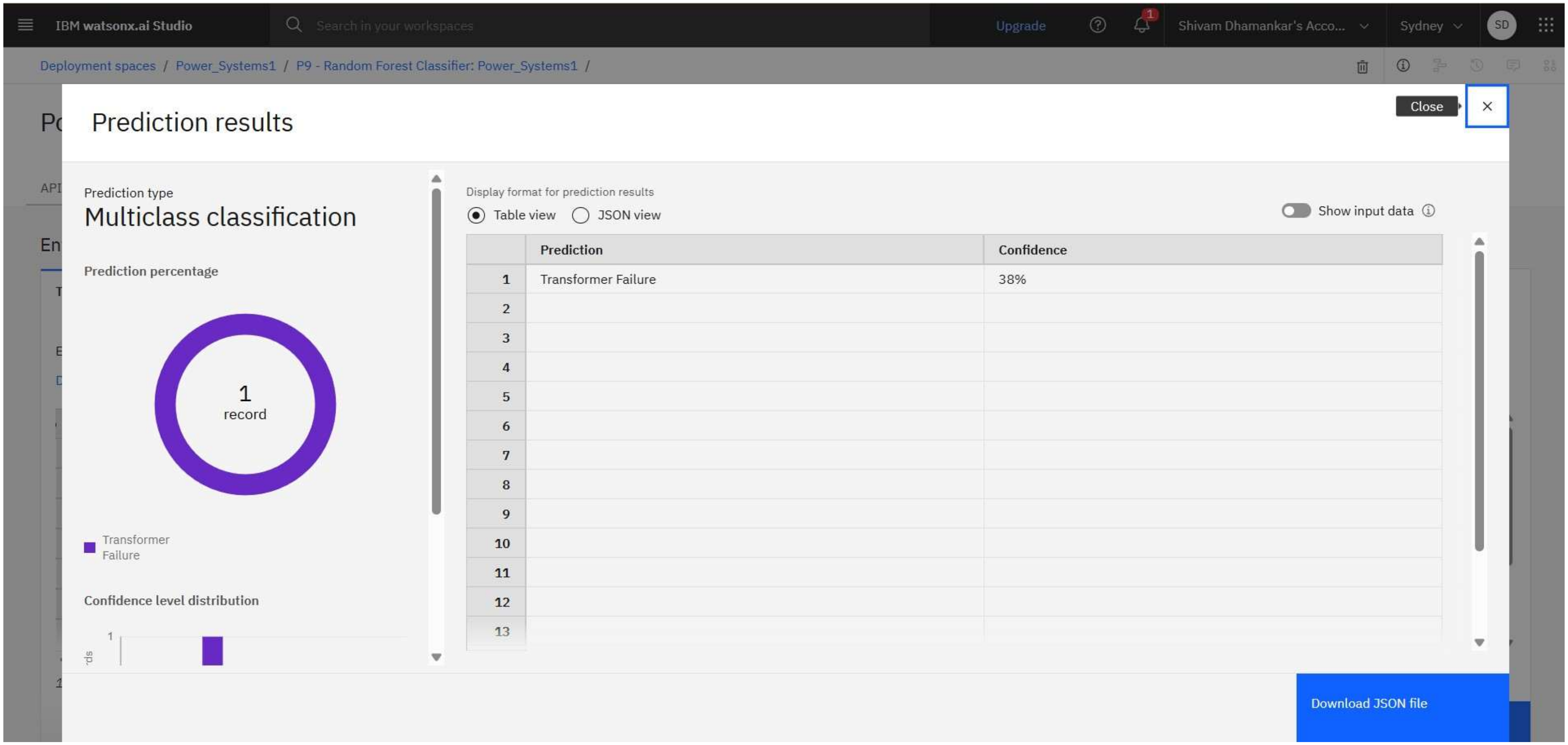
SD
Deployment spaces / Power_Systems1 / P9 - Random Forest Classifier: Power_Systems1 /
Power_System1
Deployed
Online
API reference
Test
Enter input data
Text
JSON
Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.
Download CSV template
Browse local files
Search in space
Clear all
Fault ID (other)
Fault Location (Latitude, Longitude) (other)
Voltage (V) (double)
Current (A) (double)
Power Load (MW) (double)
Temperature (°C) (double)
Wind Speed (km/h) (double)
1
F015
(34.2256, -118.9178)
1848
231
49
39
13
Rainy
2
3
4
5
6
7
1 row, 12 columns
Predict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Sydney
SD
Deployment spaces / Power_Systems1 / P9 - Random Forest Classifier: Power_Systems1 /
Close
Prediction results
Prediction type
Multiclass classification
Prediction percentage
1
record
Transformer Failure
Confidence level distribution
Display format for prediction results
Table view
JSON view
Show input data
Prediction
Confidence
1
Transformer Failure
38%
2
3
4
5
6
7
8
9
10
11
12
13
Download JSON file

Novelty and Uniqueness

Novelty

The proposed system combines **Machine Learning and IBM Cloud technologies** to provide intelligent and automated fault detection in power distribution networks. Unlike traditional protection systems that rely on fixed thresholds and manual analysis, the proposed solution learns from historical electrical data and accurately identifies different fault conditions in real time. The integration of cloud-based analytics enables scalable, remote, and efficient monitoring of power system health.

Uniqueness

- Uses **Machine Learning algorithms** for automatic fault detection and classification.
- Distinguishes between multiple fault types (LG, LL, LLG, and LLL) with high accuracy.
- Deploys the model on **IBM Cloud** for scalable and real-time operation.
- Reduces dependency on manual monitoring and conventional rule-based methods.
- Provides rapid fault identification, minimizing outage duration and equipment damage.
- Supports smart grid applications through intelligent decision-making.
- Enables remote access, monitoring, and management via cloud services.
- Improves power system reliability, stability, and maintenance efficiency.

Git Hub Link

Git Hub URL - https://github.com/Shivam200408/Power-System-Fault-Detection-and-Classification-1.git

Make sure you have Uploaded following three files into your github repository

1) Python file .pynb

2) Screenshots of Result

3) CSV File

4) ReadMe File

5) PPT of the Project

6) PDF File

Future Scope

The future scope of this project involves enhancing the Power System Fault Detection and Classification system using advanced IBM Cloud services. Real-time data can be collected from IoT sensors and smart meters through IBM Cloud-based IoT platforms for continuous monitoring. Advanced AI and Deep Learning models can be developed in IBM Watson Studio to improve fault prediction accuracy. The system can be extended to support predictive maintenance by identifying potential faults before they occur. Integration with smart grid infrastructure can enable automatic fault isolation and service restoration. Additional IBM Cloud analytics and visualization services can provide real-time dashboards and insights for power system operators. The solution can also be scaled to monitor large power distribution networks and renewable energy systems, making the power grid more intelligent, reliable, and efficient.

Thank You!

Thank you for your time and interest.